

SQL Cheat Sheet

Basic Queries

Select all columns:

```
SELECT * FROM employees;
```

Select specific columns:

```
SELECT first_name, last_name, salary FROM employees;
```

Filter by condition:

```
SELECT * FROM employees WHERE salary > 8000;
```

Filter by multiple conditions:

```
SELECT * FROM employees WHERE department_id = 50 OR department_id = 80;
```

Filter by pattern matching (LIKE):

```
SELECT * FROM employees WHERE first_name LIKE 'M%';
```

Sort results:

```
SELECT first_name, last_name, salary FROM employees ORDER BY salary DESC;
```

Limit results:

```
SELECT * FROM employees FETCH FIRST 10 ROWS ONLY;
```

String Functions

Concatenate columns:

```
SELECT first_name || ' ' || last_name AS "Full Name" FROM employees;
```

Convert to uppercase/lowercase:

```
SELECT UPPER(first_name), LOWER(last_name) FROM employees;
```

Find string length:

```
SELECT LENGTH(first_name) FROM employees;
```

Find position of a character:

```
SELECT INSTR(last_name, 'a') FROM employees;
```

Extract part of a string:

```
SELECT SUBSTR(first_name, 1, 2) FROM employees;
```

Numeric Functions

Power function:

```
SELECT POWER(4, 2) FROM dual; -- Output: 16
```

Square root:

```
SELECT SQRT(121) FROM dual; -- Output: 11
```

Ceiling value:

SELECT CEIL(3.5) FROM dual; -- Output: 4

Floor value:

SELECT FLOOR(-3.2) FROM dual; -- Output: -4

Modulo (remainder):

SELECT MOD(10, 3) FROM dual; -- Output: 1

Round number:

SELECT ROUND(745.126, 2) FROM dual; -- Output: 745.13

Truncate number:

SELECT TRUNC(745.126, 2) FROM dual; -- Output: 745.12

Absolute value:

SELECT ABS(-2.3) FROM dual; -- Output: 2.3

Character Functions

Convert to lowercase:

SELECT LOWER('HELLO') FROM dual; -- Output: hello

Convert to uppercase:

SELECT UPPER('hello') FROM dual; -- Output: HELLO

Capitalize first letter:

SELECT INITCAP('hello world') FROM dual; -- Output: Hello World

Concatenation:

SELECT CONCAT('Hello', ' World') FROM dual; -- Output: Hello World

Substring extraction:

SELECT SUBSTR('Hello', 1, 3) FROM dual; -- Output: Hel

Find string length:

SELECT LENGTH('Hello') FROM dual; -- Output: 5

Find position of character:

SELECT INSTR('Hello', 'l') FROM dual; -- Output: 3

Left pad string:

SELECT LPAD('Hello', 10, '*') FROM dual; -- Output: *****Hello

Right pad string:

SELECT RPAD('Hello', 10, '*') FROM dual; -- Output: Hello*****

Trim characters:

SELECT TRIM(LEADING 'H' FROM 'Hello') FROM dual; -- Output: ello

Replace characters:

SELECT REPLACE('Hello World', 'World', 'SQL') FROM dual; -- Output: Hello SQL

Joins

Employees and their department:

```
SELECT first_name, last_name, department_id FROM employees  
WHERE department_id IN (SELECT department_id FROM departments);
```

Employees and their managers:

```
SELECT first_name, last_name, manager_id FROM employees  
WHERE manager_id IN (SELECT manager_id FROM employees);
```

Salary Updates**Increase salary by 10%:**

```
SELECT first_name, last_name, salary, salary + salary * 0.10 FROM employees;
```

Decrease salary by 200:

```
SELECT first_name, last_name, salary, salary - 200 FROM employees;
```